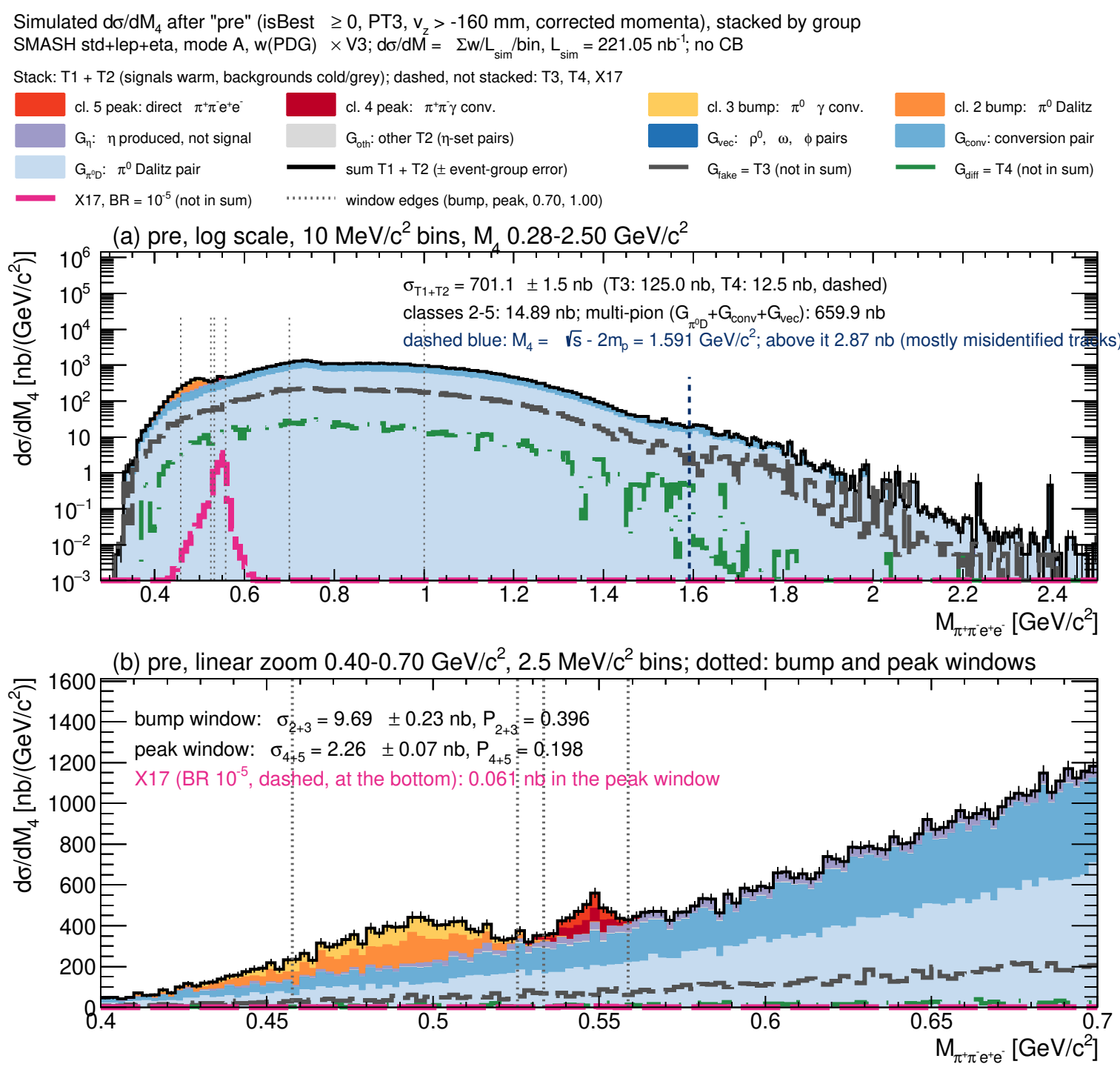
Simulated dσ/dM4 after "pre" (isBest ≥ 0, PT3, vz > -160 mm, corrected momenta), stacked by group
SMASH std+lep+eta, mode A, w(PDG) × V3; dσ/dM = Σw/Lsim/bin, Lsim = 221.05 nb-1; no CB
Stack: T1 + T2 (signals warm, backgrounds cold/grey); dashed, not stacked: T3, T4, X17
cl. 5 peak: direct π+π-e+e-
cl. 4 peak: π+π-γ conv.
cl. 3 bump: π0 γ conv.
cl. 2 bump: π0 Dalitz
Gη: η produced, not signal
Goth: other T2 (η-set pairs)
Gvec: ρ0, ω, ϕ pairs
Gconv: conversion pair
Gπ0D: π0 Dalitz pair
sum T1 + T2 (± event-group error)
Gfake = T3 (not in sum)
Gdiff = T4 (not in sum)
X17, BR = 10-5 (not in sum)
window edges (bump, peak, 0.70, 1.00)
(a) pre, log scale, 10 MeV/c2 bins, M4 0.28-2.50 GeV/c2
σT1+T2 = 701.1 ± 1.5 nb (T3: 125.0 nb, T4: 12.5 nb, dashed)
classes 2-5: 14.89 nb; multi-pion (Gπ0D+Gconv+Gvec): 659.9 nb
dashed blue: M4 = √s - 2mp = 1.591 GeV/c2; above it 2.87 nb (mostly misidentified tracks)
dσ/dM4 [nb/(GeV/c2)]
Mπ+π-e+e- [GeV/c2]
(b) pre, linear zoom 0.40-0.70 GeV/c2, 2.5 MeV/c2 bins; dotted: bump and peak windows
bump window: σ2+3 = 9.69 ± 0.23 nb, P2+3 = 0.396
peak window: σ4+5 = 2.26 ± 0.07 nb, P4+5 = 0.198
X17 (BR 10-5, dashed, at the bottom): 0.061 nb in the peak window
dσ/dM4 [nb/(GeV/c2)]
Mπ+π-e+e- [GeV/c2]